APT-BENCH: PATIENT-DISJOINT EVALUATION OF GRANULARITY AND TWO-AXIS SCALING IN SINGLE-CELL APTAMER PROFILING

Anonymous authors
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ABSTRACT

Single-cell assays produce hundreds of thousands of cells, but evaluation on new subjects is determined by the number of independent patients. We introduce APT-Bench, a patient-disjoint benchmark of 361,792 single-cell aptamer profiles from 40 patients across six disease groups. It separates five-lineage and 27-subtype cell annotation from patient-level characterization and makes both patient diversity and within-patient cell depth explicit training axes. Cell-level splitting inflates performance, and across neural and classical models coarse lineage recognition remains substantially easier than fine subtype prediction. At each of three fixed total-cell budgets, reallocating training cells from 8 to 32 patients improves macro-F1 for both Logistic Regression and XGBoost at both resolutions; matched doublings and a descriptive response surface likewise favor broader patient coverage over the evaluated range. A strict nested bridge aggregates disease-label-free XGBoost subtype probabilities without exposing either stage to outer-test patients. Predicted subtype composition retains disease-associated variation (patient macro-F1 0.593), but its gains over lineage composition and a cell-count-only control have patient-bootstrap intervals spanning zero. Stronger direct APT discrimination is sensitive to patient-specific absolute-expression signatures whose biological and technical origins cannot be separated with the available metadata. APT-Bench therefore provides a statistically explicit setting for studying observational unit, annotation granularity, training scale, and cross-task error propagation, while limiting clinical conclusions to this cohort pending external validation.

1 INTRODUCTION

Cancer screening and diagnosis are usually framed at the patient level: does a person have disease, and if so which disease category is most likely? Single-cell liquid biopsy changes this framing. A single blood draw can support both a patient-level diagnostic decision and a cell-level characterization of the circulating immune compartment. These outputs are related, but they do not live at the same observational unit. Disease is a property of the patient, whereas lineage and subtype are properties of the cell.

This mismatch matters for evaluation. Our benchmark contains 361,792 single cells, but only 40 patients. Treating cells as independent disease observations would substantially overstate the effective sample size of the diagnostic task. It would also hide a second asymmetry: lineage and subtype form a genuine cell-level hierarchy, but disease does not sit above that hierarchy. The clinically relevant problem is therefore not a single three-level taxonomy. It is a multi-resolution setting in which different targets require different evaluation protocols, inductive biases, and interpretations.

We study aptamer-based single-cell profiling. Apt-seq uses sequencing-compatible nucleic-acid affinity reagents to quantify molecules such as surface proteins while retaining cell barcodes and paired transcriptomic state (Wu et al., 2025). Unlike gene-vocabulary scRNA-seq or antibody-derived-tag integration benchmarks, our predictive input is a fixed 293-aptamer binding panel. This enables three questions: how performance changes from coarse lineage to fine subtype; how patient-level disease evaluation handles many correlated cells but only 40 subjects; and how many measured aptamers and held-out cells preserve that disease signal at inference.

APT-Bench fixes three patient-disjoint tracks: coarse-to-fine cell typing, patient-level disease recognition with statistical-unit audits, and measurement efficiency over aptamer-measurement and inference-aggregation budgets. Its reusable contribution is not a claim that one cohort spans deployment, but a common contract for a new modality: immutable patient folds, explicit observational units, leakage-resistant metrics, and outer-test-isolated budget controls that can extend to future clinical single-cell cohorts.

To instantiate the hierarchy track, we use DropCascade, a differentiable lineage-subtype cascade. The model combines a shared APT encoder, soft lineage routing, and subtype experts. We compare it with classical models and encoder-matched flat and HCE-trained Transformer controls. It serves as a hierarchy-aware neural reference rather than a claimed universal winner; the matched controls test whether any gain survives control of encoder capacity and training budget.

This perspective leads to three empirical findings. First, coarse lineage recognition is substantially easier than fine subtype recognition across all models, and encoder-matched controls do not support a DropCascade-specific advantage. Second, a controlled two-axis scaling study separates patient diversity from within-patient cell depth. At each fixed total training-cell budget, reallocating cells from 8 to 32 patients improves both coarse and fine macro-F1 for Logistic Regression and XGBoost; averaged matched doublings and descriptive response surfaces also favor the patient axis over the evaluated range. Effect magnitude still depends on model, resolution, and budget. This separates two claims that are often conflated: patient-disjoint splits are necessary for valid generalization estimates, while correlated cells can still be useful training observations. Third, strictly nested, disease-label-free XGBoost subtype probabilities retain disease-associated variation after patient aggregation: soft subtype composition reaches disease macro-F1 0.593 versus 0.477 for lineage composition. The paired difference is uncertain, however, and subtype composition does not significantly exceed a cell-count-only control. The bridge is therefore predictive and associational; direct APT discrimination is stronger but sensitive to patient-specific absolute-expression structure.

Our contributions are therefore as follows.

- **APT-Bench, a patient-disjoint benchmark for single-cell APT liquid biopsy.** We formalize three tasks with fixed patient folds, common inputs and labels, and a portable evaluation contract spanning hierarchy prediction, patient-level disease audit, and measurement efficiency.
- **A multiresolution account of the granularity gap.** Beyond aggregate macro-F1, we report joint hierarchy metrics, subtype-level support analyses, and patient-clustered comparisons under a common evaluation protocol. We use DropCascade as a hierarchy-aware reference rather than claiming architectural superiority.
- **Controlled two-axis scaling and downstream error propagation.** Fixed-total-cell and matched-doubling experiments distinguish patient diversity from within-patient depth, while strict nested cross-fitting converts disease-label-free subtype probabilities into patient compositions without exposing either stage to outer-test patients.

2 Related Work

Liquid biopsy and multi-cancer detection. Blood-based multi-cancer early detection has focused primarily on circulating tumor DNA, methylation, fragmentomics, and bulk protein assays (Cohen et al., 2018; Zeng et al., 2026; Wan et al., 2025; Kahwati et al., 2025). These approaches target a patient-level diagnostic decision. Our setting differs because APT profiling observes individual circulating cells, which introduces a second prediction problem at cell resolution: identifying lineage and fine subtype structure within the same blood draw. The resulting task is not simply "more labels," but a change in statistical unit and hierarchy structure.

Single-cell foundation models and modality mismatch. Large pretrained models such as Geneformer, scGPT, and scFoundation learn representations from transcriptomic data and are widely used as reference points for single-cell representation learning (Theodoris et al., 2023; Cui et al., 2024; Hao et al., 2024). Their success does not transfer automatically to our setting. The input modality is different, since APT consists of a fixed 293-dimensional protein-oriented panel rather than a gene-vocabulary sequence. More importantly, recent evaluation work shows that frozen or zero-shot single-cell foundation-model embeddings are often matched by simpler baselines once the downstream task and evaluation protocol are made explicit (Kedzierska et al., 2025). This motivates training directly on APT profiles and treating evaluation design itself as a first-class contribution.

Reusable single-cell benchmarks. Open Problems separates datasets, methods, and metrics behind public loaders and evaluation code (Luecken et al., 2025); a 2025 multimodal benchmark similarly releases processed RNA/ADT/ATAC inputs and task-specific pipelines (Liu et al., 2025). SPARE standardizes sample-representation interfaces (Shitov et al., 2025), and PaSCient evaluates patient representations with study-separated partitions (Liu et al., 2026). None combines a fixed aptamer-binding panel, coarse-to-fine cell typing, patient-level disease audit, and aptamer-measurement/inference-aggregation budgets. APT-Bench targets this narrower gap rather than replacing broad RNA or integration benchmarks.

Hierarchical cell-type classification. Cell identity naturally admits coarse-to-fine structure, and recent work has explored losses or architectures that exploit this structure for annotation and out-of-distribution robustness (Jia et al., 2025; Cultrera di Montesano et al., 2026). Our use of hierarchy is narrower and more explicit: disease is kept outside the hierarchy, while lineage and subtype form the only genuine taxonomy. We compare DropCascade not only with flat classifiers but also with a matched flat FT-style Transformer control (Gorishniy et al., 2021) and the same control trained with a two-level adaptation of hierarchical cross-entropy (HCE), the most directly related hierarchy-aware loss.

Measurement-budget sensitivity. Clinical translation of high-plex assays depends not only on accuracy but also on measurement cost. As a secondary sensitivity, APT-Bench asks how many aptamers are needed under outer-test-isolated panel selection and how many held-out-cell predictions suffice for patient aggregation. The latter is an inference sensitivity, not reduced-cell training. This is distinct from biomarker discovery: our goal is to quantify predictive budget and redundancy in the assay, not to claim that a small panel is a universal causal signature.

Tabular deep learning versus classical machine learning. APT expression is a medium-dimensional tabular signal. In that regime, gradient-boosted trees remain strong defaults and often rival or outperform deep tabular models (Grinsztajn et al., 2022; Erickson et al., 2026). This literature is directly reflected in our results: XGBoost and encoder-matched Transformer controls remain competitive at both resolutions, so DropCascade cannot be justified by a blanket accuracy claim. We instead use it as a structured reference for the hierarchy diagnostics, while disease recognition is audited with simple, strong tabular baselines.

3 Problem Setup

3.1 Dataset and Prediction Targets

Cohort construction and specimen source. APT-Bench is derived from peripheral-blood clinical specimens collected from 40 participants: patients with colorectal cancer (CRC), gastric cancer (GC), hepatocellular carcinoma (HCC), pancreatic cancer (PC), or biliary tract cancer (BTC), and healthy controls (Normal). The six groups contain 6–8 participants each. Clinical specimens were processed into single-cell suspensions before assay preparation. The study workflow documents quality control using the forward/side-scatter cell profile and viable-cell proportion before library construction.

Paired single-cell assay and label construction. The assay jointly measures transcriptomic state and binding of 293 target-specific nucleic-acid aptamers on the same cells. Transcriptome and aptamer-barcode libraries are constructed separately and linked at the cell level by their shared barcodes. Transcriptomic counts were processed and clustered with the standard Seurat workflow (Hao et al., 2021); predicted doublets were removed with DoubletFinder (McGinnis et al., 2019), and low-quality cells were filtered using per-cell UMI count, detected-gene count, and mitochondrial-transcript fraction. Investigators then manually annotated clusters from canonical subtype marker expression reported in prior PBMC and pan-cancer studies (Zhang et al., 2024; Yang et al., 2024). These transcriptome-derived annotations define the lineage and subtype targets, while disease labels come from participant-level cohort assignment. The benchmark input contains only the 293 APT measurements; transcriptomic measurements construct the reference labels but are never exposed to a prediction model. Exact parameters and label limitations are documented in Appendix C.1.

TODO: Complete the remaining cohort, specimen-processing, assay, and annotation parameters listed in Appendix C.1; report unavailable metadata explicitly rather than leaving them implicit.

Benchmark cohort. The linked post-QC cohort contains 374,854 cells. After excluding the transcriptome-derived *Unknown* subtype label, which denotes uncertain annotation rather than a validated biological state, the main benchmark contains 361,792 cells. Each cell is represented by a 293-dimensional APT expression vector. Expression values are log-transformed and standardized feature-wise using statistics estimated from the training partition only. The 293 aptamers are designed probes rather than an arbitrary learned basis.

APT-Bench contains two kinds of targets at different observational units.

- **Patient-level disease recognition.** Each patient is assigned to one of six disease groups.
- **Cell-level hierarchy prediction.** Each cell is assigned to one of five coarse immune lineages and one of 27 fine subtypes. Every subtype maps to exactly one parent lineage through a manually curated lookup table. The same subtype names and operational definitions are used across all six disease groups rather than being redefined by disease.

Disease is therefore not treated as the root of a three-level hierarchy. It is a separate patient-level label observed on the same blood draw.

TODO: Document independent biological review of the subtype-to-lineage lookup, including reviewers, disagreements, and adjudication.

3.2 Benchmark Contract and Extensibility

APT-Bench fixes the unit of input, prediction target, split, and metric for each track: five-lineage and 27-subtype prediction from one APT vector per cell; two-axis scaling over training patients and cells per patient; and strict cross-fitted aggregation from cell predictions to six-class patient-level characterization with gold-composition and technical controls. Novelty is not part of the primary contract. Outer-test-isolated aptamer measurement and inference-cell budgets are secondary sensitivities. The machine-readable manifest at `benchmark/splits/patient_folds.json` assigns every pseudonymous patient to one outer fold; all selection remains inside development patients. Portable components include patient-clustered uncertainty, hierarchy diagnostics, pseudoreplication and fingerprint audits, and outer-test-isolated budget curves. The full interface and release contract appear in Appendix C.2. This cohort cannot estimate cross-site or cross-platform generalization.

TODO: Deposit the versioned data and executable evaluation release specified in Appendix C.2; otherwise describe the work as a dataset case study rather than a reusable benchmark.

3.3 Evaluation Protocols

We evaluate models under two main protocols with different purposes.

Patient-level 5-fold cross-validation. This is the primary protocol for all headline claims. Patients are partitioned into five disease-stratified folds, each containing eight held-out test patients. At every fold, preprocessing statistics, model fitting, and hyperparameter selection use only the training and validation patients from the remaining folds. We pool out-of-fold predictions across all five test folds, yielding a patient-disjoint evaluation spanning all 40 patients.

Cell-level split control. As a protocol-sensitivity control, cells are randomly split irrespective of patient identity. Cells from the same patient can therefore appear in both training and evaluation sets. We do not treat this as a realistic deployment protocol. Its only role is to quantify performance inflation caused by leakage.

Training patient–cell scaling. The original 4×4 surface is retained as a descriptive sensitivity, but its unequal endpoints do not compare patient and cell effects. The primary fair design uses $P \in \{8, 16, 32\}$ outer-development patients and $C \in \{100, 200, 400, 800, 1600\}$ uniformly sampled cells per patient. Its fixed-total diagonals hold $T = PC \in \{3200, 6400, 12800\}$ exactly constant, and the full grid also supports matched doublings of one axis while holding the other fixed. Within every outer fold, 20 disease-stratified patient orderings and uniform within-patient cell permutations are nested across budgets. Preprocessing is fit only on sampled training cells; frozen Logistic Regression and XGBoost configurations are refit for coarse and fine prediction; and every cell from the same eight untouched outer-test patients is evaluated. We report pooled cell-weighted macro-F1 and macro-F1 computed within each test patient and then averaged. Empirical subset-seed intervals and paired patient-clustered bootstrap intervals answer different uncertainty questions and are reported separately. This training-time design is distinct from inference subsampling in the compact-panel analysis.

3.4 Class Distribution

Because the subtype distribution is long-tailed, we use macro-F1 as the primary metric for coarse-lineage and fine-subtype prediction. Accuracy is secondary. Primary point estimates pool cells and are therefore cell-weighted despite patient-disjoint splits; Appendix C.5 reports equal-cell-per-patient sensitivity results. For uncertainty, we compute 95% confidence intervals from 2,000 paired patient-level bootstrap resamples over the 40 pooled test patients. The primary inferential family contains DropCascade versus HCE and versus XGBoost at the coarse and fine endpoints. Two-sided bootstrap p-values use twice the smaller tail probability with a plus-one correction and are Holm-adjusted across these four tests; comparisons outside this family are exploratory.

From the same pooled predictions, joint hierarchy metrics are exact-path accuracy, root-excluded example-averaged lineage/subtype F1, and subtype-tree distance; support correlations, sibling errors, and the coarse–fine gap are exploratory difficulty diagnostics.

We report disease recognition at the patient level, since the clinical decision is made per patient rather than per cell. Cell-level disease accuracies are included only as supplementary evidence and are not interpreted as though hundreds of thousands of cells were independent disease observations.

4 Models and Nested Evaluation

APT-Bench is an evaluation contribution rather than a claim of architectural superiority. Classical controls include class-weighted multinomial Logistic Regression and XGBoost. The neural reference, DropCascade, uses a shared four-layer APT Transformer, an auxiliary five-lineage head, a global subtype path, soft-routed lineage experts, and a cross-disease supervised contrastive term. Encoder-matched controls remove the cascade and use either independent lineage/subtype cross-entropy or HCE (Cultrera di Montesano et al., 2026). All models use the same patient folds and training-only preprocessing; neural checkpoints are selected by validation fine macro-F1. Complete architectures, objectives, and optimization settings are given in Appendix B.

For the composition bridge, each disease outer fold seals eight patients. A final disease-label-free XGBoost subtype model trained on the other 32 patients constructs outer-test compositions. Downstream training compositions for those 32 patients are instead produced by four inner patient-disjoint fits, each trained on 24 patients. A class-weighted multinomial LR, with regularization selected by patient-level inner CV, is then evaluated once on the sealed eight patients. Thus neither the cell-typing stage nor the disease stage uses an outer-test patient during preprocessing, feature construction, fitting, or selection. Appendix B.4 provides the complete data flow and controls.

5 Results

5.1 Patient-Disjoint Hierarchy Prediction Exposes a Granularity Gap

Coarse lineage recognition is consistently easier than fine subtype recognition (Table 1). For DropCascade, coarse macro-F1 is 0.326 while fine macro-F1 is 0.152; the matched flat FT-style

Model	Coarse M-F1	Fine M-F1	Fine Acc.	Exact Path	Root-excluded Hier. F1	Subtype Tree Dist.↓
DropCascade	0.326	**0.152**	0.305	0.253	0.404	2.431
Flat FT-style Transformer	0.323	0.147	0.315	0.257	0.411	2.407
FT-style Transformer + HCE	**0.333**$^{\text{n.s.}}$	0.151$^{\text{n.s.}}$	0.306	**0.259**	0.405	2.418
XGBoost	0.322$^{\text{n.s.}}$	0.143$^{\text{n.s.}}$	**0.323**	0.248	**0.420**	**2.375**

Table 1: Patient-disjoint 5-fold hierarchy evaluation on the same pooled out-of-fold predictions from 40 patients. Exact Path and Root-excluded Hier. F1 use the independent coarse and fine predictions. Exact Path requires both predictions to be correct. Root-excluded Hier. F1 is the example-averaged F1 between the two-node true and predicted sets $\{\text{lineage}, \text{subtype}\}$; we exclude the always-correct root node to avoid inflating every model through a trivial shared ancestor. Subtype Tree Dist. compares the true and predicted subtypes through their mapped parents and does not use the separate coarse prediction: 0 for the correct subtype, 2 for a sibling, and 4 for a cross-lineage prediction (lower is better). No metric uses constrained decoding. The primary comparisons are DropCascade versus HCE and versus XGBoost for coarse and fine macro-F1; their two-sided tests use 2,000 paired patient bootstrap resamples with Holm correction across four tests. Other comparisons and the three hierarchy metrics are descriptive. $^{\text{n.s.}}$ denotes no Holm-adjusted significance; coarse–fine consistency remains in Appendix C.9.

Transformer reaches 0.323 and 0.147. The corresponding gap appears across model families, indicating that APT profiles recover broad immune identity more reliably than the finer transcriptome-derived taxonomy.

Encoder-matched comparisons do not establish an architectural advantage. Relative to the flat Transformer, DropCascade differs by only $+0.0036$ coarse macro-F1 (paired patient-bootstrap 95% CI $[-0.0078, 0.0146]$) and $+0.0055$ fine macro-F1 (95% CI $[-0.0021, 0.0125]$). HCE has the highest coarse point estimate (0.333), but DropCascade$-$HCE is -0.0069 (95% CI $[-0.0154, 0.0026]$) at the coarse level and $+0.0015$ (95% CI $[-0.0083, 0.0047]$) at the fine level. Differences from XGBoost also do not exclude zero. Exact-path accuracy is 0.248–0.259; HCE has the highest value, whereas XGBoost has the highest root-excluded hierarchical F1 (0.420) and lowest subtype-tree distance (2.375). Natural consistency is reported only as an appendix diagnostic because constrained decoding can alter it without improving correctness.

5.2 Why 361,792 Cells Still Yield Only 0.15 Subtype Macro-F1

Per-subtype F1 is more strongly associated with cell abundance than with patient coverage in this cohort. Marginal $\text{Spearman}(F_{1,k}, \log n_k)$ versus $\text{Spearman}(F_{1,k}, P_k)$ is 0.486/0.288 for DropCascade and 0.540/0.275 for XGBoost. After rank residualization, $\rho_{c|p}$ is 0.411/0.496 and $\rho_{p|c}$ is $-0.051/-0.130$. Because there are only 27 subtypes and 16 lie at the 40-patient coverage ceiling, these are exploratory associations rather than causal explanations.

Only 25.0% of DropCascade errors and 24.6% of XGBoost errors retain the true parent. The raw cross-lineage rate depends on the available wrong-class distribution: against an error-conditioned marginal null, the observed rates are close to expectation and do not independently demonstrate either hierarchy preservation or failure. Appendix C.4 reports per-subtype patient-clustered intervals, support plots, and the complete error analysis. Across models, the descriptive coarse-minus-fine macro-F1 gap ranges from 0.099 to 0.200.

5.3 Patient Disjointness Changes the Evaluation, Not Just the Split

Cell-level splitting inflates coarse/fine macro-F1 by $+0.090/+0.108$ for DropCascade and $+0.050/+0.069$ for XGBoost (Appendix C.5). APT-Bench therefore fixes patient-disjoint evaluation rather than treating the split as a reporting detail. Because a patient-disjoint pooled cell metric can still overweight patients with more cells, an equal-cell sensitivity also samples 1,900 cells per patient over 50 seeds. It changes coarse macro-F1 by at most $+0.005$ and fine macro-F1 by -0.007 to -0.009, leaving the qualitative ordering intact.

5.4 Patient Diversity and Cell Depth Are Separate Scaling Axes

The fair scaling experiment holds the outer-test patients fixed and varies only the development set. It evaluates exact fixed-total-cell reallocations, matched doublings of either axis, and a fold-adjusted descriptive response surface for Logistic Regression and XGBoost at both resolutions. Every point uses 20 matched disease-stratified patient/cell subset seeds; preprocessing and model fitting are repeated on the sampled training set, while all cells from the eight untouched patients in each outer fold are evaluated.

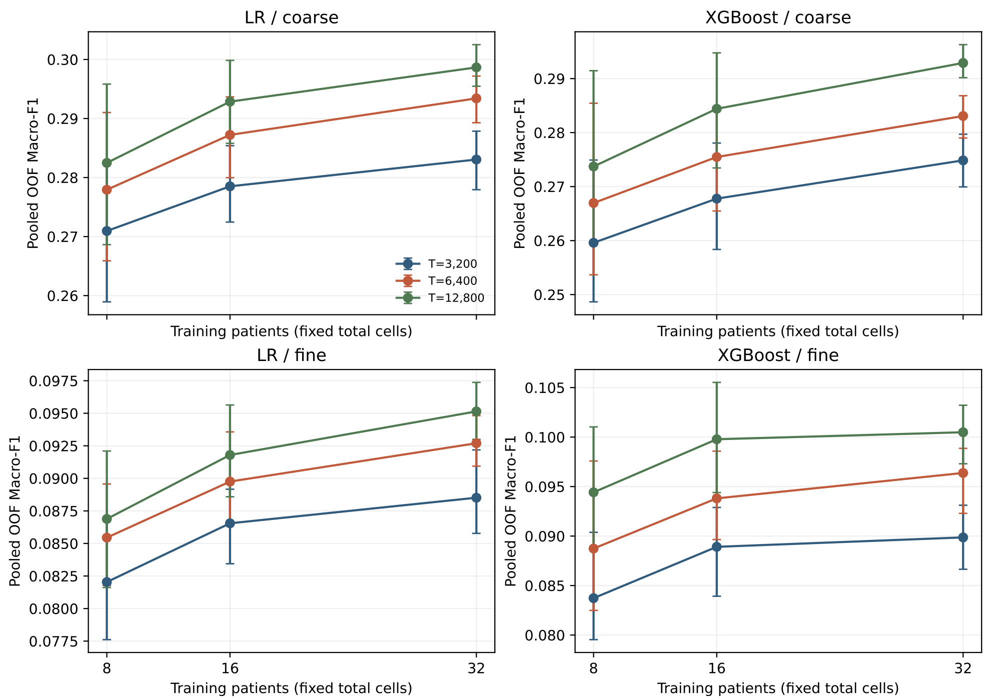

Figure 1: Fixed-total training-cell scaling. Each curve holds total sampled training cells T constant while reallocating them across P independent patients, so cells per patient equal T/P. Points are pooled patient-disjoint OOF macro-F1 means over 20 matched subset seeds; bars are empirical 95% seed intervals, not confidence intervals over patients. Models evaluate all cells from the same untouched outer-test patients. Patient-clustered intervals and patient-balanced sensitivities are reported in Appendix C.6.

At each fixed total of 3,200, 6,400, or 12,800 training cells, reallocating cells from 8 to 32 patients improves pooled OOF macro-F1 in all 12 model–resolution–budget combinations (Figure 1). The gains are 0.012–0.016 for LR coarse, 0.006–0.008 for LR fine, 0.015–0.019 for XGBoost coarse, and 0.006–0.008 for XGBoost fine. All 12 paired patient-bootstrap intervals are above zero; empirical subset-seed intervals are above zero in 8 of 12 comparisons, reflecting a separate source of variation. Mean within-patient macro-F1 gives the same direction except that XGBoost fine effects remain seed-uncertain.

Matched doublings lead to the same descriptive ordering. Averaged across eligible fixed settings, patient-versus-cell doubling gains are 0.011 versus 0.005 (LR coarse), 0.007 versus 0.003 (LR fine), 0.016 versus 0.008 (XGBoost coarse), and 0.009 versus 0.006 (XGBoost fine). Fold-adjusted response-surface coefficients are likewise larger for $\log_2 P$ than $\log_2 C$ in all four comparisons, although interactions show that neither effect is constant across the grid. Thus, within this cohort and budget range, deeper sampling from fewer patients does not compensate for broader patient coverage. This is an observational scaling result, not a universal or causal law.

5.5 Nested Composition Preserves Disease-Associated Variation

Composition connects cell typing to patient characterization, but a pooled OOF pipeline is insufficient: a cell model generating a downstream training patient's features can have seen the current disease outer-test patients. We therefore seal each eight-patient outer fold, construct the 32 development compositions by four-way inner patient-disjoint cross-fitting, and use one cell model fitted on all 32 development patients only for the sealed test patients.

Under this strict protocol, disease-label-free XGBoost soft subtype composition reaches patient accuracy/macro-F1/AUROC $0.600/0.593/0.854$, compared with $0.475/0.477/0.766$ for soft lineage composition. Both exceed their complete 10,000-permutation nulls ($p = 0.00010$ and $p = 0.00020$). Fine granularity is not established as superior: the paired subtype-minus-lineage macro-F1 difference is $+0.116$ with patient-bootstrap 95% CI $[-0.015, 0.262]$. True subtype composition reaches macro-F1 0.516, so the predicted representation is not merely a noisier estimate of annotated abundance; it may also retain model confidence or technical structure.

Log cell count alone reaches macro-F1 0.531 ($p = 0.00010$), and predicted subtype composition exceeds it by only $+0.062$ (95% CI $[-0.087, 0.218]$). Equal-cell sensitivity, a global-multinomial composition null, and a fold-ID prediction control do not reveal a cell-count or fold signature sufficient to explain the complete bridge, but they also do not establish biological mediation. The defensible finding is that imperfect patient-disjoint cell predictions preserve patient-level disease-associated variation; incremental value over coarse or technical summaries remains unresolved.

Aggregation / Representation	Patient Acc.	Macro-F1	Macro-AUROC	Perm. p
Cell-level classifiers*, majority vote	**1.000**	**1.000**	–	–
Patient-level APT median + LR†	0.950	0.948	1.000	0.0010
Log cell count + LR‡	0.575	0.531	0.800	0.00010
Nested soft subtype composition + LR§	0.600	0.593	0.854	0.00010
Nested soft lineage composition + LR§	0.475	0.477	0.766	0.00020
Gold subtype composition + LR‡	0.500	0.516	0.776	0.00010
Gold lineage composition + LR‡	0.325	0.301	0.743	0.0184

* Five cell-level classifiers share the $40/40$ majority-vote result; their distinct cell metrics appear in Table 4. † One 293-APT median vector per patient. ‡ Patient-level technical or paired-transcriptomic oracle features. § Soft probabilities from disease-label-free XGBoost under strict nested cross-fitting. Composition rows use raw proportions; LR regularization is tuned only within each outer-development set. Composition permutation tests use 10,000 patient-label shuffles.

Table 2: **Patient-level** disease classification ($n = 40$). The majority-vote accuracy has 95% Clopper–Pearson CI $[0.912, 1.000]$. Predicted composition retains disease-associated variation, but its gains over lineage composition and cell count have bootstrap intervals spanning zero. All disease results are within-cohort estimates, not evidence of clinical generalization; see Appendix C.3.

Direct APT discrimination is strong within this cohort but confounding-sensitive. Five cell-level baselines yield the same $40/40$ patient majority-vote result, and a patient-level median-APT model reaches accuracy 0.950. Patient-wise centering reduces majority-vote accuracy from 1.000 to 0.650, while a 40-way patient-identity diagnostic reaches 88.4% accuracy on held-out cells. Because acquisition-batch metadata and an external APT cohort are unavailable, these offsets cannot be separated into biological and technical sources. We therefore do not interpret this task as clinical validation. Compact-panel and inference-cell sensitivities are secondary analyses in Appendix C.10.

6 Discussion

APT-Bench is primarily an evaluation contribution rather than a claim of model superiority. Across matched neural and classical models, broad lineage is more reliably recovered than transcriptome-derived subtype, and no DropCascade-specific advantage is statistically supported. The resulting granularity gap and the uncertain subtype-versus-lineage gain in the strict nested composition bridge show that additional model complexity alone does not resolve fine cell typing or establish that finer predictions improve patient-level characterization.

The benchmark also separates the evaluation unit from the training-scaling axes. Although it contains 361,792 cells, performance on unseen subjects must still be estimated from only 40 independent patients; cell-level splitting inflates scores and changes apparent model differences. The fixed-total design compares broader patient coverage with deeper sampling without changing the total number of training cells, while matched doublings and a response surface expose dependence on the evaluated model, resolution, and budget. Broader patient coverage improves all fixed-total comparisons, and its mean matched doubling effect exceeds the cell-depth effect for both classical models at both resolutions. These analyses are descriptive rather than causal or universal scaling laws. Correlated cells cannot create evidence about new patients, even when additional cells improve estimation of cell-type decision boundaries. Clinical single-cell benchmarks should therefore report patient-disjoint generalization together with both patient and training-cell budgets rather than using raw cell count as effective sample size.

The disease analyses delimit what can be concluded from this cohort. A disease-label-free subtype predictor preserves disease-associated variation after patient aggregation, but its gain over lineage composition or cell count alone is not statistically resolved. Strong centering sensitivity indicates that patient-specific absolute-expression structure contributes substantially to within-cohort discrimination, while the compact-panel results establish measurement efficiency and redundancy rather than causal biomarkers. The cohort is single-site, lacks acquisition-batch metadata, and has no external validation; the LODO analysis is descriptive and cross-disease generalization remains unresolved. The fixed schema, patient splits, tasks, and evaluator are reusable in future APT and clinical single-cell cohorts (Luecken et al., 2025; Liu et al., 2025), but batch-aware, cross-site validation is required before clinical claims.

7 Conclusion

APT-Bench establishes a patient-disjoint evaluation contract for hierarchical cell typing and patient-level characterization from single-cell aptamer profiles. Across matched neural and classical models, coarse lineage is consistently easier than fine transcriptome-derived subtype prediction, and no DropCascade-specific advantage is statistically supported. Controlled fixed-total and matched-scale experiments show that broader patient coverage is the stronger scaling axis over the tested range rather than conflating either axis with raw cell count. Strict nested composition analysis further shows that imperfect cell typing can preserve patient-level disease-associated variation, although its incremental value over coarser or technical summaries is unresolved. Within-cohort disease discrimination is strongly associated with patient-specific absolute-expression structure. These findings position APT-Bench as a benchmark for statistical-unit, granularity, scaling, and error propagation effects, while external cohorts remain necessary for clinical claims.

AI use statement

TODO: **Required for ICLR 2027.** Disclose every generative-AI tool and model used for writing, editing, coding, analysis, literature review, or figure preparation; identify the tasks for which each tool was used, describe human verification of AI-assisted outputs, and state that the authors take responsibility for the final text, claims, code, and artifacts.

Ethics statement

TODO: State the human-subjects approval or exemption and informed-consent status; describe data de-identification, access restrictions, privacy protections, intended clinical/research use, foreseeable misuse or bias risks, and any conflicts of interest. Do not include identifying institutional details if they would break anonymity.

Reproducibility statement

TODO: Provide the anonymous code and data-access locations, exact patient-split manifest, preprocessing and evaluation commands, software environment, random seeds, model hyperparameters, compute requirements, and paths for the prediction artifacts and tables used in every reported result.

Author Contributions

TODO: Add author contributions using CRediT roles (for example: conceptualization, methodology, software, validation, formal analysis, data curation, visualization, writing, supervision, and funding acquisition). Preserve anonymity in the review version if required.

Acknowledgments

TODO: Add funding sources and grant numbers, computational resources, data or clinical contributors, and non-author assistance; verify whether any acknowledgment must be omitted or anonymized during double-blind review.

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A Appendix

B Hierarchy-Aware Reference Model

APT-Bench is intentionally broader than any single model. We nevertheless need a hierarchy-aware reference architecture that can operate directly on APT profiles under the benchmark protocols. DropCascade plays this role. It focuses on the genuine cell-level hierarchy, lineage $\to$ subtype, and leaves disease recognition to separate tabular baselines and patient-level aggregation. We do not present the architecture as a primary contribution: without full five-fold component ablations, its global path, coarse auxiliary objective, soft routing, and contrastive construction should be read as implementation choices rather than independently validated sources of performance.

B.1 Cell Encoder

Each cell is represented by a standardized APT vector $x \in \mathbb{R}^{293}$. We embed feature identity and feature value separately. For aptamer index i, we form a token

$$t_i = \mathrm{Emb}(i) + \mathrm{MLP}_{\mathrm{val}}(x_i), \qquad i = 1, \dots, 293. \tag{1}$$

A learnable `[CLS]` token is prepended to the resulting sequence and processed by a pre-norm Transformer encoder with 4 layers, 8 attention heads, hidden dimension 256, feed-forward dimension 512, GELU activations, and dropout 0.1. The final `[CLS]` representation

$$z = \mathrm{Encode}(x) \in \mathbb{R}^{256} \tag{2}$$

serves as the shared cell representation for downstream tasks.

B.2 Soft-Routed Hierarchical Cascade

Let $M = 5$ be the number of coarse lineages and $K = 27$ the number of fine subtypes. Each subtype k belongs to exactly one parent lineage $m(k)$. A coarse head produces lineage logits

$$c = W_c z \in \mathbb{R}^M, \tag{3}$$

with routing distribution

$$p(m \mid x) = \mathrm{softmax}(c/\tau)_m, \tag{4}$$

where $\tau = 1$.

Instead of a hard cascade, which would force subtype prediction to follow only the top predicted lineage, DropCascade uses a soft mixture of a global subtype path and lineage-specific experts:

$$s_k(x) = \underbrace{g_k(z)}_{\text{global path}} + \alpha\, p(m(k) \mid x)\, e_{m(k),k}(z) + \gamma \log\big(\epsilon + (1-\epsilon) p(m(k) \mid x)\big), \tag{5}$$

Here $g(z)$ is a global subtype head, $e_{m,k}(z)$ is the expert associated with lineage m, $\alpha = 1.0$, $\epsilon = 0.1$, and γ is ramped from 0 to 0.5 during early training. The global path retains a direct subtype-logit contribution alongside the routed expert contribution; we do not isolate its effect from the other components. Final predictions are

$$\hat{k} = \arg\max_k s_k(x), \qquad \hat{m} = \arg\max_m c_m. \tag{6}$$

B.3 Cross-Disease Contrastive Regularization

The reference model includes a cross-disease supervised contrastive term. Let

$$u = \frac{\mathrm{Proj}(z)}{\|\mathrm{Proj}(z)\|_2} \tag{7}$$

be a normalized projection of the encoder representation. For anchor cell i, positives are cells with the same subtype but a different disease label. Same-subtype cells from the same disease are excluded by construction. Cells from different subtypes are treated as negatives, with additional emphasis on negatives that share the same coarse lineage. The contrastive loss weight is ramped from 0 to 0.2 after the early warmup period. Because we do not compare this construction with standard supervised contrastive learning across all five folds, we make no claim that the cross-disease positive rule independently improves generalization.

B.4 Optimization and Model Scope

The classification cascade is trained jointly using

$$\mathcal{L} = \lambda_{\text{fine}}\mathcal{L}_{\text{fine}} + \lambda_{\text{coarse}}\mathcal{L}_{\text{coarse}} + w_{\text{contrast}}(t)\mathcal{L}_{\text{contrast}}, \tag{8}$$

where $\mathcal{L}_{\text{fine}}$ and $\mathcal{L}_{\text{coarse}}$ are cross-entropy losses with $\lambda_{\text{fine}} = 1.0$ and $\lambda_{\text{coarse}} = 0.5$. The contrastive weight increases to 0.2 according to the schedule in §B.3.

We optimize the cascade using AdamW with learning rate 10^{-4} and weight decay 10^{-5}, gradient clipping at norm 1.0, batch size 512, and mixed-precision training. Models are trained for at most 50 epochs with early stopping based on validation fine-level macro-F1 with patience 10. The final configuration does not use class-balanced sampling because it underperformed natural sampling after matching the number of optimization steps; the single-split development diagnostic is provided in Appendix C.11.

Matched Transformer controls. To isolate encoder capacity from hierarchical modeling, a flat FT-style Transformer control (Gorishniy et al., 2021) removes the cascade while retaining the identical APT tokenizer, Transformer dimensions, optimizer, early stopping, and training budget above. We call it FT-style because its tokenizer is exactly the APT feature tokenizer used by DropCascade, rather than the original numerical/categorical FT-Transformer tokenizer. This model and its HCE counterpart predict the same $M + K$ lineage-and-subtype nodes. The flat model applies independent cross-entropy to the lineage and subtype logit slices. For HCE (Cultrera di Montesano et al., 2026), let $R_{ij} = 1$ when node j is node i or its descendant, $p = \mathrm{softmax}(q)$ over all nodes, and $s = Rp$. We minimize $-\lambda_{\text{fine}} \log s_{M+k} - \lambda_{\text{coarse}} \log s_m$ for observed subtype k and lineage m. Because every fine label here is a leaf, the fine HCE term alone would reduce to ordinary CE; including the simultaneously observed parent label makes this a stated two-level adaptation rather than an unqualified reproduction of the mixed-granularity setting in the original work. All three Transformer models are evaluated from the checkpoint selected by patient-validation fine macro-F1, without an additional post-selection calibration stage.

Disease recognition is not handled by a dedicated head inside DropCascade. This is a deliberate modeling decision rather than a claim that disease is solved in a strong external sense: within-cohort disease prediction is already nearly saturated by simple classical baselines, while the architectural challenge addressed by DropCascade is the unresolved cell-level hierarchy problem.

Strictly nested composition bridge. For each disease outer fold f, we train a disease-label-free XGBoost subtype classifier on all 32 outer-development patients and predict the cells of the eight untouched patients in f. Disease-training features require a second cross-fit: for each of the other four fixed folds g, a separate subtype model is trained on the 24 patients outside $f \cup g$ and predicts fold g. We average the resulting 27-class probability vectors within patient to obtain soft subtype composition; soft lineage composition sums subtype probabilities under the frozen subtype-to-lineage map. A standardized, class-weighted multinomial LR is fit to the 32 inner-OOF patient vectors, with $C \in \{0.01, 0.1, 1, 10, 100\}$ selected by patient-level inner CV, and evaluated once on the eight outer-test vectors. Raw proportions are primary; CLR with a prespecified 10^{-6} pseudocount is a sensitivity analysis. All uncertainty and permutation analyses resample or shuffle patients, never cells.

C Dataset and Evaluation Details

C.1 Dataset Construction and Documentation Checklist

The available study workflow establishes the high-level sequence used to construct APT-Bench: clinical peripheral-blood specimens from five cancer groups and healthy controls are converted to single-cell suspensions; cell-profile and viability quality control precedes a paired single-cell transcriptome and 293-aptamer assay; separate transcriptome and aptamer-barcode libraries are linked at the cell level; and transcriptome-derived clusters and annotations provide the cell labels. For a benchmark release, this overview must be accompanied by the following provenance and protocol fields.

Transcriptomic preprocessing and cell quality control. The transcriptomic count matrix was processed with Seurat following its standard workflow of normalization, feature selection, dimensionality reduction, neighborhood-graph construction, and clustering (Hao et al., 2021). DoubletFinder was used to identify and remove predicted doublets (McGinnis et al., 2019). Low-quality cells were additionally filtered using per-cell UMI count, detected-gene count, and mitochondrial-transcript fraction. These steps preceded cluster annotation and construction of the linked APT benchmark cohort. TODO: Report the Seurat and DoubletFinder versions, every parameter and random seed, the exact UMI/gene/mitochondrial thresholds, and cell counts before and after each filter.

Manual annotation and reference-label reliability. After Seurat clustering, investigators manually assigned lineage and subtype names by inspecting canonical marker-gene expression reported in prior PBMC and pan-cancer studies (Zhang et al., 2024; Yang et al., 2024). No quantitative annotation confidence, inter-annotator agreement, or external reference-transfer score is available. Accordingly, these labels are operational reference annotations, not error-free biological ground truth, and benchmark scores are conditional on this manual taxonomy. The *Unknown* label was assigned when a cluster did not show a coherent canonical marker pattern for the expected PBMC subtypes; this was a manual reject decision rather than a calibrated confidence threshold. Unknown cells are excluded from the primary 27-subtype benchmark. TODO: Provide the full subtype-by-marker citation table, the number and expertise of annotators, whether they were blinded to disease, the adjudication procedure, and the explicit rule used to assign Unknown.

Cross-disease semantics and hierarchy mapping. The same 27 subtype labels and definitions were applied across all six disease groups; no subtype was renamed or redefined for a particular disease. This gives the labels a consistent operational meaning, but it does not prove that marker expression or biological state is invariant across diseases. Each subtype is assigned to one of five parent lineages by a manually curated lookup. TODO: Document independent validation of the subtype-to-lineage mapping, including reviewers, disagreements, adjudication, and any marker-based checks.

- TODO: Report recruiting site(s) and dates, prospective versus retrospective collection, exact patient count per group, inclusion/exclusion and diagnostic criteria, and whether each participant contributed exactly one specimen.
- TODO: Report available demographics and clinical covariates, including age, sex, disease stage, treatment status, and comorbidities; summarize missingness and state which fields cannot be released.
- TODO: Report blood volume, collection tubes, time to processing, cell-isolation and cryopreservation procedures, quality-control thresholds, and sample/cell exclusion counts.
- TODO: Report assay/platform and reagent versions, sequencing instruments and depth, 293-probe panel design, and a verified protein-target manifest or explicitly state why only anonymized probe identifiers can be released.
- TODO: Report barcode-matching rates, APT normalization and filtering, all remaining software versions and parameters, and whether annotation was performed before investigators examined APT profiles or disease labels.
- TODO: Add a cohort-flow diagram with participant and cell counts after every exclusion step, and provide the ethics approval/consent, de-identification, data-access, license, and intended-use terms referenced in the Ethics and Reproducibility Statements.

C.2 Benchmark Interface and Release Contract

The hierarchy submission interface contains pseudonymous cell and patient IDs, true and predicted lineage, and true and predicted subtype. The disease interface contains one row per patient with the true disease and six class scores. The efficiency track additionally records the aptamer measurement budget, inference-time cell budget, selection seed, and the development-only feature ranking. These common interfaces permit new models to be scored without reproducing their internal training pipelines.

The evaluation layer is portable to other clinical single-cell datasets. It implements group-disjoint out-of-fold prediction; patient-clustered bootstrap uncertainty; coarse and fine macro-F1, exact-path

Resource	Primary input	Reported scope	Main unit	Hierarchy	Patient task	Two-axis scale	Assay budget	Public pipeline
Open Problems	Multiple	12 living tasks	Task-specific	Task-specific	Task-specific	No	No	Yes
scMultiBench	RNA/ADT/ATAC	64 real + 22 simulated	Cell/dataset	No	No	No	Feature selection	Yes
SPARE	scRNA-seq	5 datasets, 8 methods	Sample	No	Yes	No	No	Yes
PaSCient	scRNA-seq	12.5M cells, 2,700 patients	Patient	No	Yes	No	No	Not audited
APT-Bench	293 aptamers	1 cohort, 40 patients	Cell + patient	Yes	Audit	Yes	Yes	Pending release

Table 3: Scope comparison with recent single-cell evaluation resources (Luecken et al., 2025; Liu et al., 2025; Shitov et al., 2025; Liu et al., 2026). "Task-specific" means that the living platform defines multiple contracts rather than one shared clinical protocol. APT-Bench is narrower in cohorts but uniquely combines a fixed aptamer modality, patient-disjoint coarse-to-fine cell typing, within-cohort patient-level audit, and controlled patient/cell and assay budgets. Its public release remains a submission requirement, not a completed claim.

accuracy, hierarchical F1, subtype-tree distance, granularity gap, and sibling versus cross-lineage error; patient-level aggregation versus cell-level pseudoreplication; patient-centering and identity diagnostics; and nested feature- and cell-budget curves. A future dataset can replace the assay features, hierarchy, or clinical endpoint while retaining these interfaces. This modular task–dataset–metric separation follows recent living single-cell benchmarks (Luecken et al., 2025); standardized metadata and programmatic access similarly enable cross-study reuse in CZ CELLxGENE (CZI Cell Science Program et al., 2025).

A complete release must contain the processed APT matrix; pseudonymous cell and patient identifiers; disease and cell labels; hierarchy and feature metadata; the immutable fold manifest; metric implementations; baseline configurations; and a single command that regenerates the headline tables from model predictions. TODO: Before submission, add data and code URLs, explicit licenses, file checksums, an environment lockfile, an archival DOI, and the end-to-end command. If participant-level data require controlled access, release the complete schema, all labels permitted by consent, the split manifest, evaluator, and access procedure.

C.3 Disease Classification: Cell-Level Source Prediction and Patient-Level Sensitivity Analyses

Cell-level prediction of the disease source of a held-out patient. Table 2 reports patient-level disease classification, the correct statistical unit for a per-patient diagnostic call. For reference and comparability with earlier reporting conventions, Table 4 additionally reports the pooled *cell-level* metric, in which every held-out cell is treated as an independent prediction target and pooled across all 361,792 cells from the five test folds. This is an easier and statistically different task than patient-level diagnosis, since cells from the same patient are correlated observations rather than independent draws and patients with more cells contribute more weight. We therefore report this metric only as a lower-level diagnostic.

Model	Accuracy	Macro-F1	Weighted-F1	Balanced Acc.
Logistic Regression	**0.991**	**0.989**	**0.991**	**0.989**
XGBoost	0.990	0.988	0.990	0.987
Linear SVM	0.988	0.986	0.988	0.985
Random Forest	0.963	0.955	0.963	0.953
Reference Correlation†	0.921	0.909	0.921	0.915

† SingleR-style reference-correlation classifier (Spearman nearest-centroid mapping), following the template-matching principle used by reference-mapping methods such as SingleR (Aran et al., 2019).

Table 4: **Cell-level** prediction of the disease source of a held-out patient's cells (6-class), pooled over all 361,792 out-of-fold cells from the patient-level 5-fold protocol. This is a different, easier statistical task than the patient-level diagnosis reported in Table 2: cells from the same patient are correlated observations rather than independent draws, and this metric implicitly weights patients with more cells more heavily. We report it here only as a cell-level reference point, not as a patient-level diagnostic result.

Patient-wise centering sensitivity. To test whether disease prediction relies on each patient's absolute expression level rather than only on within-patient relative structure, we repeated Logistic Regression training and evaluation after subtracting each patient's own per-aptamer median from every cell in both training and test data. This transformation preserves within-patient cell-to-cell variation while removing a patient-specific location shift. Cell-level accuracy drops from 0.991 to 0.401 (macro-F1 $0.989 \rightarrow 0.348$), and patient-level majority-vote accuracy drops from $40/40$ to $26/40$. This collapse shows that disease recognition depends substantially on patient-level absolute-expression structure, but it does not by itself determine whether that structure is biological, technical, or a mixture of both.

Patient-fingerprint diagnostic. We additionally trained a 40-way classifier to predict patient identity from individual cells, using a within-patient 50/50 cell split for this diagnostic only. The classifier identifies the correct patient from a single held-out cell with 88.4% accuracy (macro-F1 0.862), against a 2.5% chance level. Individual cells therefore carry a strong and easily recoverable patient-specific signature, consistent with the centering sensitivity above.

Patient-level summary-feature control. Finally, to confirm that the disease signal does not rely only on cell-level pseudo-replication, we trained a genuinely patient-level classifier using each patient's median 293-dimensional APT profile. This summary-feature model reaches patient accuracy 0.950, patient macro-F1 0.948, and macro-AUROC 1.000. Together with the sensitivity analyses above, this indicates that disease information is present at the patient level but strongly entangled with absolute-expression offsets that cannot be cleanly separated without acquisition-batch metadata.

Strict nested composition bridge. Ordinary pooled OOF predictions are insufficient for stacking because a base model used to create a disease-training feature may have seen the eventual disease outer-test patients. We therefore use the nested protocol defined in Methods: within each outer fold, four XGBoost models trained on 24 patients generate inner-OOF features for the 32 disease-training patients, and a fifth model trained on all 32 predicts the eight outer-test patients. XGBoost receives subtype labels but no disease labels. Soft subtype composition reaches accuracy/macro-F1/AUROC $0.600/0.593/0.854$ and soft lineage composition reaches $0.475/0.477/0.766$. The paired macro-F1 difference is $+0.116$ (95% patient-bootstrap CI $[-0.015, 0.262]$). Gold subtype and lineage compositions reach macro-F1 0.516 and 0.301, respectively. The apparently stronger predicted than gold point estimate is not statistically resolved and warns that predicted probabilities may encode model-confidence or technical structure beyond true cell abundance.

Across 10,000 complete patient-label permutations, predicted subtype and lineage composition have plus-one $p = 0.00010$ and $p = 0.00020$. However, log cell count alone reaches macro-F1 0.531 ($p = 0.00010$), and the predicted-subtype minus cell-count difference is $+0.062$ (95% CI $[-0.087, 0.218]$). Equal-cell subsampling at 100, 500, and 2,000 cells per patient preserves the qualitative result over 50 seeds. A 50-repeat global-multinomial composition control averages macro-F1 0.170, and fold-ID prediction from global OOF composition is below chance. These controls support a predictive and associational bridge, not causal mediation or independent biological validation.

Representation	Legacy pooled OOF	Strict nested
Gold subtype composition	0.572	0.516
Predicted subtype composition	0.400	0.593
Gold lineage composition	0.377	0.301
Predicted lineage composition	0.282	0.477

Table 5: Protocol audit of patient macro-F1. The legacy predicted rows use hard labels from one global pooled-OOF prediction set, CLR features, and fixed LR regularization; the strict rows use nested soft probabilities, raw proportions, and development-only LR selection. Because both feature construction and downstream fitting changed, the numerical differences are not estimates of a leakage effect. Only strict nested results support current claims.

Patient-label permutation control. We evaluated the same patient-level median-profile classifier under 1,000 label permutations. Disease labels were permuted across the 40 patients as indivisible units, preserving the disease-group counts $(6, 8, 7, 7, 6, 6)$; labels were never shuffled across cells. We retained the original five patient folds rather than restratifying after permutation. No permuted dataset reached the observed accuracy of 0.950 or macro-F1 of 0.948, giving plus-one permutation $p = (0 + 1)/(1000 + 1) = 0.000999$ for both statistics. The null accuracy mean was 0.137, with empirical 95% interval $[0.025, 0.275]$. This rejects a random patient-label explanation under the fixed protocol, but does not distinguish biological signal from patient-associated technical structure.

C.4 Per-Subtype Support and Performance Audit

Figure 2 visualizes the two marginal support–F1 associations for DropCascade and XGBoost. The vertical stack at 40 patients makes the coverage ceiling explicit. These plots and the partial correlations are exploratory: they neither establish causality nor test whether two dependent correlations differ significantly.

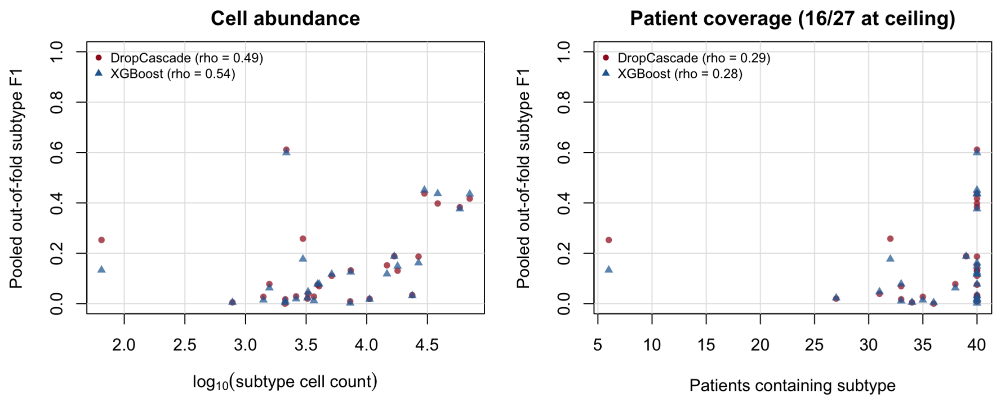

Figure 2: Per-subtype pooled out-of-fold F1 versus cell abundance (left) and patient coverage (right). Each marker is one of 27 subtypes; both models use the same ground-truth support values. Spearman correlations are marginal, and 16 of 27 subtypes occur in all 40 patients.

Model	ρ_c	ρ_p	$\rho_{c\|p}$	$\rho_{p\|c}$	G	R_{sibling}	**Cross-lineage**
DropCascade	0.486	0.288	0.411	-0.051	0.174	0.250	0.750
Flat FT-style Transformer	0.505	0.255	0.464	-0.123	0.176	0.243	0.757
FT-style Transformer + HCE	0.513	0.318	0.427	-0.034	0.183	0.257	0.743
XGBoost	0.540	0.275	0.496	-0.130	0.179	0.246	0.754
Logistic Regression	0.435	0.107	0.488	-0.267	0.167	0.295	0.705
Linear SVM	0.339	-0.012	0.463	-0.335	0.200	0.265	0.735
Random Forest	0.661	0.418	0.564	-0.035	0.158	0.191	0.809
Reference Correlation	0.376	0.003	0.499	-0.354	0.099	0.225	0.775

Table 6: Exploratory subtype diagnostics on pooled patient-disjoint predictions. ρ_c and ρ_p are marginal Spearman associations of subtype F1 with log cell count and patient coverage; $\rho_{c|p}$ and $\rho_{p|c}$ are partial Spearman correlations computed by residualizing rank-transformed variables. $G = F_1^{\text{coarse}} - F_1^{\text{fine}}$. Among errors, R_{sibling} retains the true parent lineage and Cross-lineage is its complement. With only 27 subtypes and limited variation in patient coverage, these associations are descriptive and do not establish causality.

Table 7 gives all subtype-level quantities and 95% percentile intervals from 2,000 patient-clustered bootstrap resamples. Counts and coverage use pooled ground truth, while F1 uses the same patient-disjoint OOF predictions as Table 1. Mean cells per patient averages only over patients in which the subtype is observed.

Table 7: Per-subtype support and pooled out-of-fold F1 with 95% percentile intervals from 2,000 patient-clustered bootstrap resamples. Cells/patient is the mean among patients in which that subtype is observed. Head and Tail denote the seven most and seven least abundant subtypes; all others are Mid.

Subtype	Lineage	Cells	Patients	Cells/patient	Regime	DropCascade F1 [95% CI]	XGB F1 [95% CI]
Erythrocyte	Other	65	6	10.8	Tail	0.253 [0.000, 0.395]	0.133 [0.000, 0.545]
Atypical Memory B	B	781	34	23.0	Tail	0.005 [0.000, 0.014]	0.005 [0.000, 0.014]
CD4 Monocyte-4	Myeloid	1,405	35	40.1	Tail	0.027 [0.006, 0.053]	0.014 [0.000, 0.033]
platelet	Other	1,574	38	41.4	Tail	0.078 [0.016, 0.125]	0.063 [0.015, 0.106]
Tcm	T	2,121	36	58.9	Tail	0.000 [0.000, 0.000]	0.004 [0.000, 0.016]
Memory B-2	B	2,130	33	64.5	Tail	0.017 [0.004, 0.033]	0.011 [0.000, 0.024]
Plasma	B	2,173	40	54.3	Tail	0.611 [0.522, 0.684]	0.599 [0.496, 0.685]
pDC	Myeloid	2,615	40	65.4	Mid	0.029 [0.010, 0.061]	0.019 [0.005, 0.040]
CD14 Monocyte-3	Myeloid	2,977	32	93.0	Mid	0.258 [0.170, 0.306]	0.177 [0.103, 0.211]
CD8 + CTL-3	T	3,244	27	120.1	Mid	0.020 [0.004, 0.035]	0.023 [0.008, 0.043]
GCB	B	3,283	31	105.9	Mid	0.039 [0.007, 0.071]	0.047 [0.001, 0.073]
Cycling T	T	3,667	40	91.7	Mid	0.028 [0.015, 0.043]	0.012 [0.003, 0.022]
CD14 Monocyte-2	Myeloid	3,931	40	98.3	Mid	0.075 [0.046, 0.102]	0.078 [0.056, 0.097]
CD8 + CTL-2	T	4,049	33	122.7	Mid	0.069 [0.017, 0.116]	0.078 [0.015, 0.113]
CD16 Monocyte-2	Myeloid	5,142	40	128.6	Mid	0.111 [0.064, 0.161]	0.117 [0.063, 0.164]
CD8 + CTL-1	T	7,303	40	182.6	Mid	0.009 [0.000, 0.024]	0.002 [0.001, 0.004]
cDC2	Myeloid	7,376	40	184.4	Mid	0.132 [0.099, 0.181]	0.125 [0.079, 0.167]
CD8 Tem-2	T	10,548	40	263.7	Mid	0.019 [0.005, 0.038]	0.017 [0.008, 0.030]
CD16 Monocyte-1	Myeloid	14,678	40	366.9	Mid	0.152 [0.111, 0.198]	0.118 [0.090, 0.150]
CD4 + Tcm-2	T	16,861	39	432.3	Mid	0.188 [0.111, 0.236]	0.188 [0.136, 0.220]
Naive T	T	17,960	40	449.0	Head	0.131 [0.067, 0.183]	0.149 [0.073, 0.199]
Memory B-1	B	23,722	40	593.0	Head	0.034 [0.020, 0.048]	0.032 [0.017, 0.047]
CD8 Tem-1	T	26,714	40	667.9	Head	0.187 [0.128, 0.250]	0.162 [0.107, 0.221]
CD14 Monocyte-1	Myeloid	29,871	40	746.8	Head	0.437 [0.378, 0.495]	0.451 [0.392, 0.507]
CD56 NK	NK	38,442	40	961.0	Head	0.398 [0.320, 0.436]	0.437 [0.373, 0.480]
CD16 NK	NK	58,507	40	1462.7	Head	0.383 [0.326, 0.434]	0.376 [0.317, 0.429]
CD4 + Tcm-1	T	70,653	40	1766.3	Head	0.417 [0.368, 0.460]	0.435 [0.393, 0.473]

For continuity with common long-tail reporting, Table 8 aggregates the seven least and seven most abundant subtypes. This summary is secondary to the per-class audit because individual subtypes can depart sharply from the abundance trend.

Model	Tail Macro-F1	Head Macro-F1
DropCascade	0.142	0.284
Flat FT-style Transformer	0.117	0.290
FT-style Transformer + HCE	0.125	0.288
XGBoost	0.119	0.292
Logistic Regression	0.114	0.197
Linear SVM	0.082	0.204
Random Forest	0.072	0.245
Reference Correlation	0.060	0.114

Table 8: Fine-subtype macro-F1 on the seven least and seven most abundant subtypes under pooled patient-disjoint evaluation. This aggregate is reported as a descriptive complement to the complete per-subtype audit in Table 7.

Cross-lineage error relative to an error-conditioned null. Table 9 uses the wrong-conditioned null as the primary reference: after excluding the true subtype, it renormalizes the predicted-subtype marginal so every randomized outcome remains an error. For the four main models, observed cross-lineage rates are 0.743–0.757 versus wrong-conditioned expectations of 0.741–0.745. The corresponding differences are only +0.002 to +0.013, and all four patient-bootstrap 95% intervals include zero. Thus the raw approximately 75% rate is largely explained by the available wrong-class distribution and does not independently demonstrate either hierarchy preservation or hierarchy failure. The unconditional permutation, which can create chance exact matches and therefore uses a different sample space, is retained only as a sensitivity analysis.

Model	Errors	Observed	Wrong-cond. null	Obs.−WC null [95% CI]	Uncond. null (sens.)
DropCascade	251,354	0.750	0.744	+0.005 [-0.025, +0.028]	0.675
Flat FT-style Transformer	247,798	0.757	0.745	+0.011 [-0.020, +0.035]	0.675
FT-style Transformer + HCE	250,944	0.743	0.741	+0.002 [-0.028, +0.023]	0.671
XGBoost	244,990	0.754	0.741	+0.013 [-0.020, +0.038]	0.672
Logistic Regression	301,566	0.705	0.752	-0.047 [-0.061, -0.036]	0.720
Linear SVM	296,904	0.735	0.774	-0.039 [-0.053, -0.028]	0.738
Random Forest	246,420	0.809	0.749	+0.059 [+0.021, +0.083]	0.685
Reference Correlation	328,596	0.775	0.785	-0.009 [-0.022, +0.003]	0.758

Table 9: Cross-lineage error relative to an error-conditioned marginal null. The primary wrong-conditioned (WC) null draws from each model's predicted-subtype marginal after excluding the true subtype and renormalizing, so every randomized outcome remains wrong. Observed−WC-null confidence intervals recompute both terms in 2,000 patient-clustered bootstrap resamples. The unconditional permutation preserves the realized true- and predicted-subtype marginals but can create chance exact matches; it is reported only as a sensitivity analysis.

C.5 Protocol Sensitivity: Patient-Level versus Cell-Level Splitting

Table 10 compares the primary patient-disjoint protocol with a leakage-prone cell-level split control. It is placed in the appendix because the central result uses patient-disjoint evaluation throughout; the control quantifies how much the easier protocol would inflate both absolute performance and the apparent advantage of a model family.

Task	Model	Patient-disjoint	Cell split	Inflation
Coarse lineage	XGBoost	0.322	0.372	+0.050
	DropCascade	0.326	0.417	+0.090
Fine subtype	XGBoost	0.143	0.212	+0.069
	DropCascade	0.152	0.260	+0.108

Table 10: Macro-F1 under the primary patient-disjoint protocol versus a leakage-prone cell-level split control. Cell-level splitting inflates performance for both methods, and does so more strongly for DropCascade than for XGBoost. This demonstrates that evaluation protocol can change both absolute performance and the apparent relative advantage of model families, which is why APT-Bench treats patient-disjoint evaluation as a requirement rather than a reporting option.

Patient-balanced cell-count sensitivity. Patient-disjoint evaluation prevents leakage but does not make a pooled cell-level point estimate patient-balanced: patient cell counts range from 1,929 to 22,627. Table 11 therefore recomputes both hierarchy endpoints after sampling 1,900 cells without replacement from every patient for each of 50 seeds. Coarse macro-F1 changes by at most $+0.005$, while fine macro-F1 changes by -0.007 to -0.009; the qualitative model ordering is unchanged. The main conclusions are therefore not driven by a few high-cell-count patients, although the headline pooled metric remains explicitly cell-weighted.

Task	Model	Pooled M-F1	Equal-cell M-F1 [95% interval]	Δ
Coarse lineage	DropCascade	0.326	0.331 [0.325, 0.337]	+0.005
	Flat FT-style Transformer	0.323	0.326 [0.322, 0.330]	+0.003
	FT-style Transformer + HCE	0.333	0.334 [0.328, 0.340]	+0.000
	XGBoost	0.322	0.324 [0.320, 0.329]	+0.002
Fine subtype	DropCascade	0.152	0.144 [0.138, 0.153]	-0.008
	Flat FT-style Transformer	0.147	0.138 [0.131, 0.147]	-0.008
	FT-style Transformer + HCE	0.151	0.142 [0.137, 0.151]	-0.009
	XGBoost	0.143	0.136 [0.131, 0.144]	-0.007

Table 11: Patient-balanced cell-count sensitivity. For each of 50 seeds, 1,900 cells are sampled without replacement from every patient, and pooled macro-F1 is recomputed on the resulting 76,000 cells. All models use identical sampled row indices. The interval is the empirical 2.5th–97.5th percentile across seeds; Δ is the equal-cell mean minus the original pooled metric. This diagnostic changes evaluation weights only and does not retrain or reselect models.

C.6 Fair Fixed-Total and Matched-Scale Analysis

The primary scaling design uses $P \in \{8, 16, 32\}$ and $C \in \{100, 200, 400, 800, 1600\}$, with 20 nested subset seeds for every outer fold. Because every retained patient has at least 1,929 cells, all finite caps are feasible without replacement. Disease-stratified patient orderings are nested across P, uniform cell permutations are nested across C, no subtype label enters sampling, and preprocessing is refit on each sampled training set. The fixed-total diagonals satisfy $PC \in \{3200, 6400, 12800\}$ exactly.

Model	Task	Total cells	$P = 8$	$P = 16$	$P = 32$
LR	Coarse	3,200	0.271 [0.259, 0.283]	0.279 [0.272, 0.285]	0.283 [0.278, 0.288]
LR	Coarse	6,400	0.278 [0.266, 0.291]	0.287 [0.280, 0.294]	0.293 [0.289, 0.297]
LR	Coarse	12,800	0.282 [0.269, 0.296]	0.293 [0.286, 0.300]	0.299 [0.295, 0.303]
LR	Fine	3,200	0.082 [0.078, 0.087]	0.087 [0.083, 0.089]	0.089 [0.086, 0.092]
LR	Fine	6,400	0.085 [0.082, 0.090]	0.090 [0.087, 0.094]	0.093 [0.091, 0.095]
LR	Fine	12,800	0.087 [0.082, 0.092]	0.092 [0.089, 0.096]	0.095 [0.093, 0.097]
XGBoost	Coarse	3,200	0.260 [0.249, 0.275]	0.268 [0.258, 0.278]	0.275 [0.270, 0.280]
XGBoost	Coarse	6,400	0.267 [0.254, 0.285]	0.275 [0.265, 0.285]	0.283 [0.279, 0.287]
XGBoost	Coarse	12,800	0.274 [0.260, 0.291]	0.284 [0.273, 0.295]	0.293 [0.290, 0.296]
XGBoost	Fine	3,200	0.084 [0.080, 0.090]	0.089 [0.084, 0.093]	0.090 [0.087, 0.093]
XGBoost	Fine	6,400	0.089 [0.082, 0.098]	0.094 [0.090, 0.099]	0.096 [0.092, 0.099]
XGBoost	Fine	12,800	0.094 [0.089, 0.101]	0.100 [0.094, 0.106]	0.100 [0.097, 0.103]

Table 12: Fixed-total training-cell scaling. Each row holds the total number of sampled training cells constant while reallocating them across independent patients. Entries are pooled patient-disjoint OOF Macro-F1 means with empirical 95% intervals over 20 matched subset seeds. All outer-test cells are evaluated.

Table 12 reports variability across subset seeds. Separately, paired patient-clustered bootstraps over the same 40 pooled OOF patients compare $P = 8$ with $P = 32$ after averaging seed-level sufficient statistics. All 12 intervals exclude zero; their lower endpoints range from 0.0002 to 0.0084 and upper endpoints from 0.0090 to 0.0314. These intervals quantify uncertainty over patients, whereas the empirical seed intervals quantify sensitivity to which development patients and cells are sampled.

Matched doubling effects average patient versus cell gains of 0.011/0.005 (LR coarse), 0.007/0.003 (LR fine), 0.016/0.008 (XGBoost coarse), and 0.009/0.006 (XGBoost fine). In the descriptive fold-fixed-effect response surface, the corresponding per-doubling β_P/β_C coefficients are 0.017/0.008, 0.008/0.004, 0.016/0.007, and 0.009/0.006. Complete pairwise effects, patient intervals, seed intervals, and response-surface uncertainty are stored in the machine-readable scaling artifacts. The coefficients are associations over this grid, not causal or universal scaling laws.

C.7 Secondary Descriptive Patient–Cell Scaling Surface

This historical 4×4 surface fixes eight outer-test patients and varies only the training set. For each outer fold and each of 20 seeds, we generate one approximately proportional disease-stratified ordering of the 32 development patients, giving nested sets $S_8 \subset S_{16} \subset S_{24} \subset S_{32}$. Each selected patient receives one uniform random cell ordering, giving nested prefixes of 100, 500, and 2,000 cells and the complete available set. Patients with fewer than a requested cap contribute all available cells. No cell is selected using its subtype, and every model is evaluated on all cells from the untouched test patients. The full-data endpoint is fitted once per fold rather than duplicated across seeds. All 6,020 model–task jobs completed, and the deterministic endpoint exactly reproduces the frozen pooled OOF baselines within absolute tolerance 5×10^{-4}. Because $P{=}8 \rightarrow 32$ is a fourfold change whereas $C{=}100 \rightarrow$ all is a larger and patient-dependent change, this surface is descriptive only. All direct comparisons between the two axes use the fixed-total and matched-doubling analysis in the main paper.

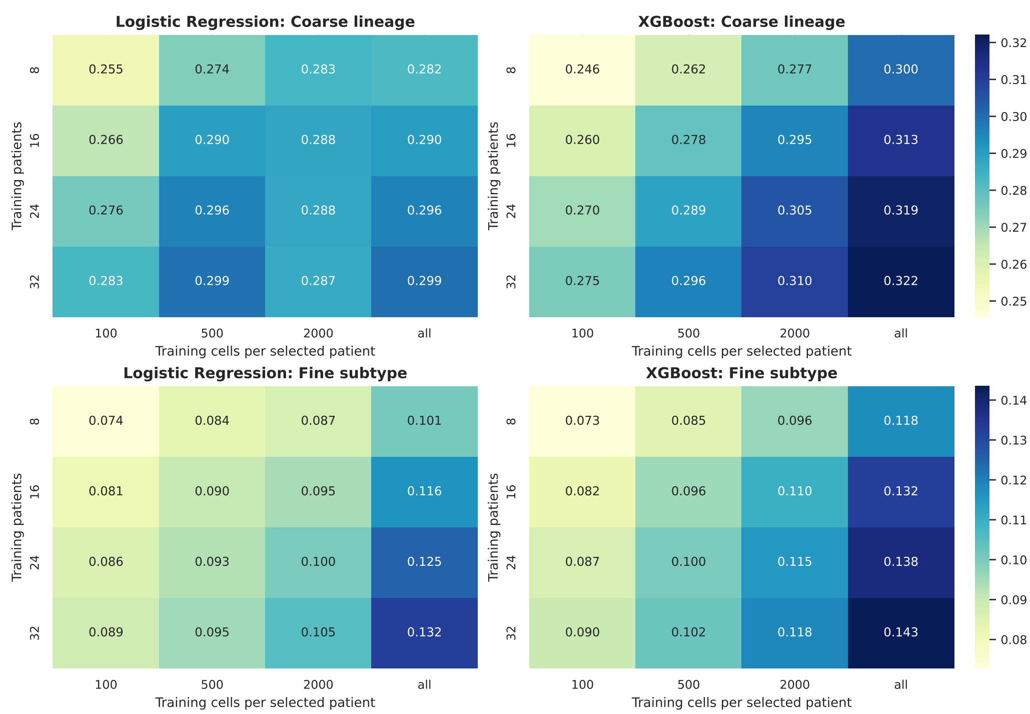

Figure 3: Pooled patient-disjoint Macro-F1 over the complete training-patient and per-patient training-cell grid. Values average 20 nested subset seeds except for the deterministic full-data endpoint. Color scales are shared across models within each task.

Model	Task	$C = 100$	$C = 500$	$C = 2000$	$C = \text{all}$
Logistic Regression, $P = 8$	Coarse	0.255 [0.247, 0.265]	0.274 [0.263, 0.284]	0.283 [0.269, 0.297]	0.282 [0.269, 0.293]
Logistic Regression, $P = 16$	Coarse	0.266 [0.261, 0.270]	0.290 [0.284, 0.297]	0.288 [0.282, 0.295]	0.290 [0.280, 0.300]
Logistic Regression, $P = 24$	Coarse	0.276 [0.269, 0.280]	0.296 [0.288, 0.304]	0.288 [0.282, 0.295]	0.296 [0.289, 0.302]
Logistic Regression, $P = 32$	Coarse	0.283 [0.278, 0.288]	0.299 [0.296, 0.302]	0.287 [0.284, 0.288]	0.299 [0.284, 0.315]
Logistic Regression, $P = 8$	Fine	0.074 [0.068, 0.081]	0.084 [0.079, 0.088]	0.087 [0.082, 0.092]	0.101 [0.091, 0.109]
Logistic Regression, $P = 16$	Fine	0.081 [0.078, 0.085]	0.090 [0.088, 0.094]	0.095 [0.090, 0.099]	0.116 [0.110, 0.125]
Logistic Regression, $P = 24$	Fine	0.086 [0.083, 0.089]	0.093 [0.089, 0.097]	0.100 [0.095, 0.105]	0.125 [0.118, 0.131]
Logistic Regression, $P = 32$	Fine	0.089 [0.086, 0.092]	0.095 [0.093, 0.097]	0.105 [0.102, 0.108]	0.132 [0.112, 0.146]
XGBoost, $P = 8$	Coarse	0.246 [0.234, 0.263]	0.262 [0.250, 0.278]	0.277 [0.262, 0.293]	0.300 [0.283, 0.321]
XGBoost, $P = 16$	Coarse	0.260 [0.252, 0.272]	0.278 [0.268, 0.288]	0.295 [0.283, 0.306]	0.313 [0.298, 0.330]
XGBoost, $P = 24$	Coarse	0.270 [0.264, 0.276]	0.289 [0.284, 0.295]	0.305 [0.297, 0.312]	0.319 [0.311, 0.328]
XGBoost, $P = 32$	Coarse	0.275 [0.270, 0.280]	0.296 [0.293, 0.298]	0.310 [0.308, 0.312]	0.322 [0.300, 0.345]
XGBoost, $P = 8$	Fine	0.073 [0.068, 0.079]	0.085 [0.080, 0.091]	0.096 [0.090, 0.103]	0.118 [0.110, 0.126]
XGBoost, $P = 16$	Fine	0.082 [0.077, 0.085]	0.096 [0.092, 0.101]	0.110 [0.103, 0.120]	0.132 [0.125, 0.140]
XGBoost, $P = 24$	Fine	0.087 [0.081, 0.092]	0.100 [0.095, 0.104]	0.115 [0.110, 0.121]	0.138 [0.132, 0.144]
XGBoost, $P = 32$	Fine	0.090 [0.087, 0.093]	0.102 [0.100, 0.105]	0.118 [0.117, 0.121]	0.143 [0.127, 0.159]

Table 13: Patient–cell training scaling. Entries are pooled OOF Macro-F1 means with empirical 2.5th–97.5th seed intervals; the deterministic $P = 32, C = \text{all}$ endpoint instead uses a patient-clustered bootstrap interval. Seed intervals are not patient confidence intervals.

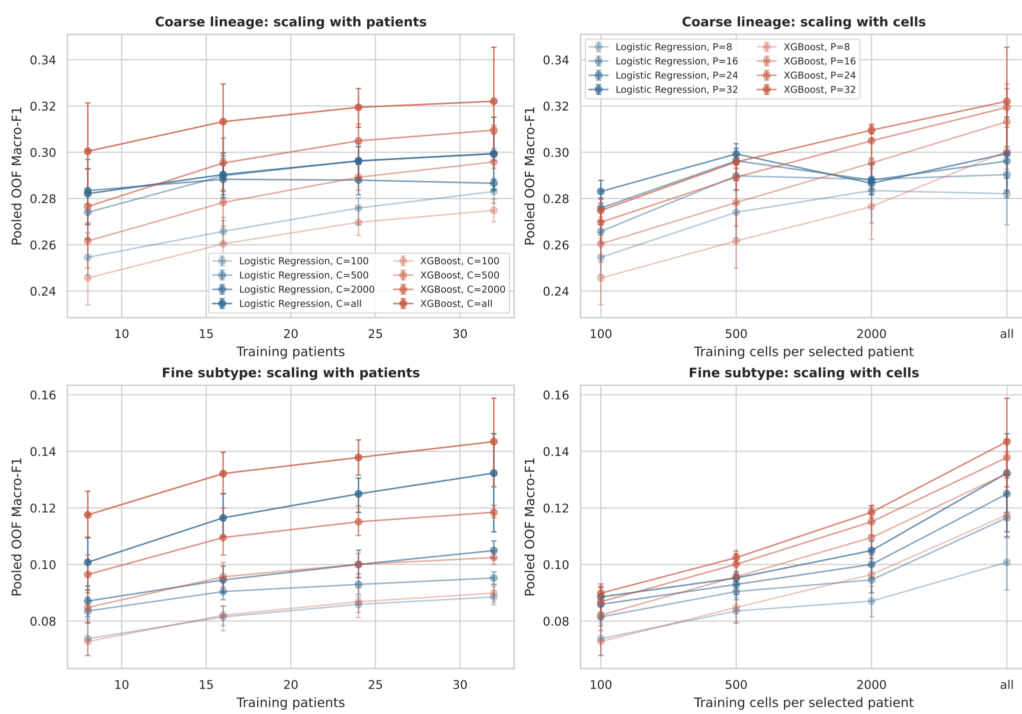

Figure 4: Patient-budget curves at each fixed per-patient cell cap. Intervals show subset-resampling variability; they do not represent uncertainty over a new patient population. Non-monotonic local changes are retained rather than smoothed.

The surface is cohort- and range-specific. We do not compare its unequal endpoints or use it to establish a patient- or cell-dominant scaling law, and it cannot estimate the benefit of patients drawn from new sites or acquisition batches.

C.8 Patient-Level Lineage-Resolved Attribution Audit

We revisit disease attribution at the patient, rather than cell, level. For patient p, lineage m, and aptamer j, we compute $\bar{x}_{pmj} = n_{pm}^{-1} \sum_{i \in (p,m)} x_{ij}$, so every point in Figures 5 and 6 represents one patient regardless of cell count. We show the three aptamers with the best mean rank across the five outer-fold PSAS rankings (APT-179, APT-225, and APT-164) in the three major immune compartments. The raw view is paired with a patient-centered view obtained by subtracting each patient's per-aptamer median across all of their cells before lineage aggregation. Disease groups contain only 6–8 patients, and the two views use the same vertical scale.

This is an exploratory full-cohort visualization, not an independently validated biomarker analysis. Raw patient means separate several disease groups, but much of that separation contracts toward zero after patient centering, consistent with the centering sensitivity in Appendix C.3. The available feature manifest contains only anonymized APT identifiers and no verified protein-target mapping. We therefore make no molecular-plausibility, protein-target, or pathway claim from these panels.

Figure 5: Raw patient-level lineage-resolved distributions for the three highest-ranked stable PSAS aptamers. Each dot is one patient-lineage mean and boxes summarize patients, not cells; disease-group sample sizes are shown on the horizontal axis. Aptamer selection and visualization use the full cohort and are exploratory.

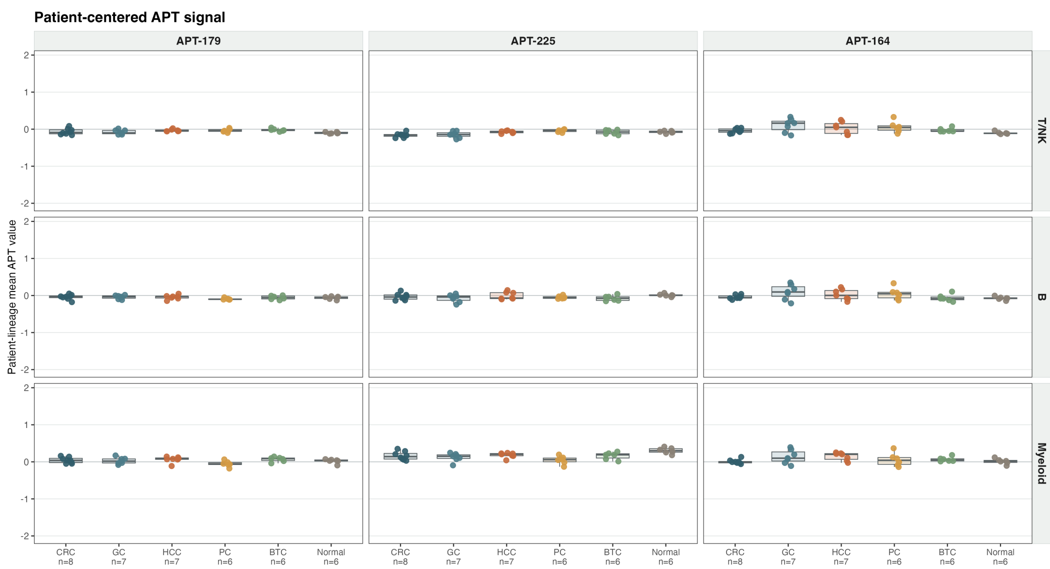

Figure 6: Patient-centered counterpart to Figure 5. Before lineage aggregation, each patient's per-aptamer median across all cells is subtracted. Each dot is one patient-lineage mean and boxes summarize patients, not cells. The vertical limits match the corresponding raw panels so that contraction after centering is directly visible. Aptamer selection and visualization use the full cohort and are exploratory.

C.9 Coarse–Fine Consistency Diagnostic

Independent coarse and fine classifiers can disagree about the hierarchy even when each prediction is plausible in isolation. Table 14 reports their natural agreement. Unlike an earlier single-split analysis, this version uses the same five-fold pooled out-of-fold prediction artifacts as the main hierarchy results, so every patient contributes exactly once to both evaluations.

Model	**Coarse–Fine Consistency**
DropCascade	0.812
Flat FT-style Transformer	0.795
FT-style Transformer + HCE	0.834
XGBoost	0.736
Logistic Regression	0.568
Linear SVM	0.631
Random Forest	0.573
Reference Correlation	0.380

Table 14: Natural coarse–fine consistency on the same 361,792-cell, 40-patient pooled out-of-fold artifacts as Table 1. For each model, the hard fine prediction is mapped to its parent and compared with the corresponding hard coarse prediction. *Unknown* is excluded. Consistency is supporting evidence rather than predictive superiority because it can be changed by constrained decoding.

C.10 Minimal Aptamer Panel for Disease Classification

To characterize assay measurement sensitivity, we asked how many of the 293 aptamers preserve the observed within-cohort disease result. This is not a clinical deployment or biomarker analysis. Patient-Stable Aptamer Selection (PSAS) denotes the fold-specific ranking and development-only budget-selection protocol below; "stable" refers to comparison of ranks across outer folds, not to patient-balanced fitting of the LR ranker.

Algorithm 1: PSAS within outer fold o.

1. Seal the eight outer-test patients; denote the other 32 patients by D_o.
2. Using all cells in D_o, fit development-only standardization and an L1-regularized six-class multinomial LR (SAGA, $C = 0.1$); rank aptamer j by $s_j = \sum_{d=1}^{6} |\beta_{dj}|$.
3. Seed-shuffle the patients in D_o into five patient-disjoint, non-stratified inner folds.
4. For every $B \in \{5, 10, 20, 40, 80, 160, 293\}$ and inner fold, refit LR on the inner-training patients using the fixed Top-B ranking and score patient-majority-vote macro-F1 on inner-validation patients.
5. Let $\bar{F}_B$ denote mean inner-validation macro-F1 and select $B^* = \min\{B : \bar{F}_B \geq \bar{F}_{293} - \delta\}$ with $\delta = 0.02$.
6. Refit LR and XGBoost from scratch on all patients in D_o, using only the LR-selected Top-B^* features, and evaluate once on the sealed outer-test patients.
7. Repeat independently for all five outer folds; report the pooled outer-test predictions.

Implementation scope and controls. The LR ranker is a cell-level disease model: cells are not first aggregated to patient medians and receive equal weight, so its coefficients are cell-count weighted rather than patient-balanced. The full-panel reference $\bar{F}_{293}$ is the mean inner-validation score, not a resubstitution estimate on all of D_o. The absolute margin $\delta = 0.02$ was fixed in code before outer-test scoring but was not prospectively preregistered. Because the ranking is computed once from all of D_o and then held fixed across its inner folds, budget selection is development-only but conditional on that ranking, not fully nested inner re-ranking.

For Random-B, each $B \in \{5, 10, 20\}$ has 100 uniformly sampled feature sets. The feature set for one repeat is shared across the five outer folds, but LR and XGBoost are trained from scratch within every fold. XGBoost in Table 15 and Figure 7 uses the LR-selected panel. The separate SHAP control instead fits full-panel XGBoost in each outer-development set, constructs a patient-balanced mean absolute TreeSHAP ranking, and refits both classifiers on SHAP-selected Top-10/20 features; it does not determine B^* or the main curve.

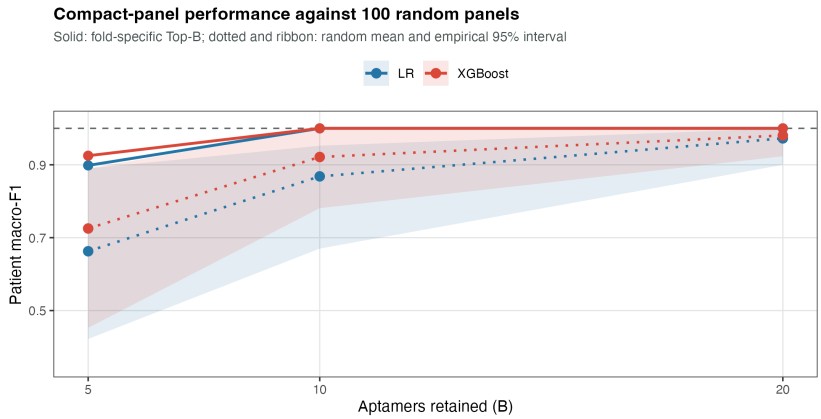

Figure 7: Patient-level disease macro-F1 under compact aptamer-budget controls. Solid lines show pooled out-of-fold performance obtained using fold-specific Top-B panels; dotted lines and ribbons show the mean and empirical 95% interval over 100 random-panel repeats. Each repeat shares one panel across outer folds, with models refit foldwise. The dashed line is the full 293-aptamer reference; scores pool the 40 outer-test patients once each.

All five outer folds select $B^* = 10$ under this rule.

Budget terminology. An *aptamer measurement budget* B means that a model is retrained and evaluated using only those B input features. An *inference-time cell budget* C means that, after training on all available outer-development cells, only C cached predictions are sampled from each held-out patient before majority-vote aggregation. We do not reduce or evaluate a *training-time cell budget*; the cell-count sensitivity therefore supports low-cell inference aggregation, not training with only C cells per patient.

The resulting development-selected 10-aptamer panels reproduce the full-panel patient-level macro-F1 of 1.0 for both Logistic Regression and XGBoost on all five outer test folds. Figure 7 compares these selected panels with 100 Random-B repeats at each compact budget. For every model and budget, we report the random-panel mean, empirical 2.5th–97.5th percentile interval, and the empirical midrank percentile of Top-B within that distribution (Table 16).

Outer Fold	δ	B^*	LR Macro-F1	XGBoost Macro-F1
0	0.02	10	1.000	1.000
1	0.02	10	1.000	1.000
2	0.02	10	1.000	1.000
3	0.02	10	1.000	1.000
4	0.02	10	1.000	1.000
All folds	**0.02**	**10**	**1.000**	**1.000**

Table 15: Outer-test-isolated patient-level panel selection for disease recognition. In each outer fold, aptamer ranking and budget selection are performed using outer-development patients only, and the selected budget is evaluated once on the untouched outer-test patients. The ranking is fixed across inner validation splits; B is selected conditional on that ranking. Under the analysis-specified, non-preregistered tolerance margin $\delta = 0.02$, all five folds select the same development budget $B^* = 10$, and the resulting 10-aptamer panels preserve the observed full-panel patient macro-F1 for both Logistic Regression and XGBoost. This result supports a measurement-efficiency interpretation rather than a claim of unique causal biomarkers.

B	Model	Top-B	Random mean [95% interval]	Percentile
5	LR	0.898	0.663 [0.422, 0.890]	97.0
5	XGBoost	0.925	0.725 [0.453, 0.890]	99.0
10	LR	1.000	0.868 [0.670, 0.952]	100.0
10	XGBoost	1.000	0.921 [0.781, 1.000]	97.0
20	LR	1.000	0.973 [0.900, 1.000]	79.5
20	XGBoost	1.000	0.981 [0.923, 1.000]	75.0

Table 16: Compact-panel random controls over 100 panel repeats. Top-B uses fold-specific outer-development LR rankings. Each repeat draws one random panel shared across outer folds; both classifiers are refitted in every fold. The percentile is the empirical midrank of Top-B within the random distribution.

At $B = 10$, the Top-B panel is at the 100th empirical percentile for LR and the 97th for XGBoost; Random-B means are 0.868 and 0.921, respectively. By $B = 20$, random means rise to 0.973 and 0.981 and many random panels also saturate, reducing the corresponding midrank percentiles to 79.5 and 75.0. Thus ranking is most consequential at the smallest budgets, while the broader panel contains extensive substitutable signal.

We next crossed aptamer budget $B \in \{5, 10, 20, 293\}$ with inference-time cell caps $C \in \{100, 500, 1000, 5000, \text{all}\}$. For each finite cap, cached predictions were sampled without replacement within each outer-test patient; patient majority votes were pooled across the five held-out folds, and the process was repeated over 50 sampling seeds. Training still used all outer-development cells. At $B = 5$, macro-F1 varies with the sampled cells and reaches 0.881 on average at $C = 100$; at $B = 10$, 20, and 293, macro-F1 is 1.0 for every cap and every seed (Table 17).

B	100 cells	500 cells	1,000 cells	5,000 cells	All cells
5	0.881 [0.845, 0.925]	0.894 [0.852, 0.925]	0.906 [0.875, 0.925]	0.899 [0.898, 0.898]	0.898 [0.898, 0.898]
10	1.000 [1.000, 1.000]	1.000 [1.000, 1.000]	1.000 [1.000, 1.000]	1.000 [1.000, 1.000]	1.000 [1.000, 1.000]
20	1.000 [1.000, 1.000]	1.000 [1.000, 1.000]	1.000 [1.000, 1.000]	1.000 [1.000, 1.000]	1.000 [1.000, 1.000]
293	1.000 [1.000, 1.000]	1.000 [1.000, 1.000]	1.000 [1.000, 1.000]	1.000 [1.000, 1.000]	1.000 [1.000, 1.000]

Table 17: LR patient-level macro-F1 under inference-time equal-cell subsampling. Entries are mean [empirical 95% interval] over 50 seeds; cached outer-test predictions are subsampled before aggregation. Models are trained once using all outer-development cells, so this is not a training-time cell-budget experiment.

Finally, we tested whether the compact result depends on Logistic Regression coefficients as the attribution mechanism. Within each outer-development set, we fitted a full-panel XGBoost model and computed absolute TreeSHAP contributions on at most 1,000 cells per development patient. We first averaged over cells within each patient and then over patients, preventing high-cell-count patients from dominating the attribution ranking. The LR–SHAP Jaccard overlap is moderate rather than complete (0.501 for Top-10 and 0.475 for Top-20), yet panels from either ranking achieve patient macro-F1 1.0 when evaluated with either LR or XGBoost (Table 18). This cross-model control supports measurement efficiency and predictive redundancy, not a unique molecular signature.

B	LR–SHAP Jaccard	Ranking	LR eval.	XGB eval.
10	0.501 ± 0.126	LR coefficient	1.000	1.000
10	–	XGBoost-SHAP	1.000	1.000
20	0.475 ± 0.087	LR coefficient	1.000	1.000
20	–	XGBoost-SHAP	1.000	1.000

Table 18: Cross-model attribution control. Jaccard is mean $\pm$ SD across five outer-fold Top-B sets. LR-coefficient and XGBoost-SHAP rows use their own selected features; evaluation columns report pooled patient macro-F1 after refitting LR or XGBoost. SHAP ranking does not select B^*.

The fold-overlap LR ranking is also stable. The highest-ranked aptamers by mean outer-fold rank are APT-179, APT-225, APT-164, APT-166, APT-57, APT-69, APT-125, APT-161, APT-172, and APT-218, and 290 of the 293 aptamers are retained after a simple redundancy screen at $|r| \leq 0.9$.

Appendix C.8 visualizes the top three at the patient-lineage level, including a patient-centering control; because only anonymized IDs are available, neither analysis supports protein-target or pathway interpretation.

C.11 Balanced Sampling Ablation

A development-only class-balanced-sampling variant did not outperform natural sampling after matching optimization steps and was not retained.

C.12 Exploratory Leave-One-Disease-Out Stress Test

We retain only a leave-one-disease-out (LODO) representation stress test. For each run, all patients from one disease category are withheld from training and used only for evaluation. Table 19 shows that generalization weakens substantially relative to the standard patient-disjoint folds, especially for fine subtypes. Because there are only six disease groups and no external cohort, LODO is descriptive rather than an estimate of deployment robustness.

Held-out Disease	Fine Macro-F1	Fine Acc.	Coarse Macro-F1	Coarse Acc.	Tail Macro-F1
BTC	0.086	0.317	0.310	0.479	0.073
CRC	0.105	0.277	0.309	0.520	0.112
GC	0.063	0.269	0.281	0.488	0.057
HCC	0.063	0.246	0.264	0.383	0.072
PC	0.057	0.253	0.221	0.482	0.037
Macro-average	**0.075**	**0.272**	**0.277**	**0.470**	**0.070**

Table 19: Leave-one-disease-out generalization for DropCascade over the five malignant held-out diseases. Each row withholds all patients of the named disease from training. The in-distribution reference using the same model under the standard patient-disjoint 5-fold protocol is fine Macro-F1 0.152 and coarse Macro-F1 0.326. A separate Normal-held-out run produces near-chance fine performance. LODO is reported only as an exploratory appendix stress test, not as proof that the model already solves cross-disease generalization.

We omit unmatched detector rankings. Cross-disease generalization remains unresolved and requires external cohorts with prespecified distribution shifts and patient-clustered uncertainty.